

Total No. of Questions : 8]

SEAT No. :

PC2383

[Total No. of Pages : 2

[6354]-500

B.E. (Computer Engineering)

NATURAL LANGUAGE PROCESSING

(2019 Pattern) (Semester- VIII) (Elective - V) (410252 A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data necessary.

- Q1)** a) Write a note on Generative Models of Language. [4]
b) Describe LDA for topic modeling with example. [8]
c) Describe HMM with help of example. [6]

OR

- Q2)** a) Write a note on Latent Semantic Analysis (LSA). [4]
b) Explain BERT. [8]
c) What is Smoothing? Explain Laplace/Add-1 smoothing with example. [6]

- Q3)** a) Define following w. r. t. Information Retrieval [6]
i) Term Frequency
ii) Inverse Document Frequency
b) Explain Information Retrieval architecture with neat diagram. [8]
c) Describe co-reference resolution with example. [4]

OR

- Q4)** a) Explain Entity Extraction and Relation Extraction w. r. t. NER. [6]
b) Explain Cross Lingual Information Retrieval with example. [8]
c) Describe Vector Space Model for Information Retrieval. [4]

P.T.O.

Q5) a) Describe Lesk algorithm and Walker's algorithm for Word Sense Disambiguation. **[10]**

b) Write a note on SpyCy Library. **[7]**

OR

Q6) a) Explain following Lexical Knowledge Networks: **[10]**

i) TreeBanks

ii) PropBanks

iii) WordNet

iv) IndoWordNet

v) VerbNets

b) List tools available for development of NLP applications. Write features of any 3 tools. **[7]**

Q7) a) Explain Rule Based Machine Translation and Statistical Machine Translation with suitable diagram and example. **[10]**

b) Explain Question Answering system with neat diagram. **[7]**

OR

Q8) a) Write a note on-Sentiment Analysis. **[10]**

b) Explain Natural Language Generation with reference architecture. **[7]**

